

Orlie Fepou

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Nationality: Cameroonian

Bochum, Germany

Personal Profile

Research intern at the Max Planck Institute for Security and Privacy, working on side-channel analysis and post-quantum cryptography. My recent MSc research focused on fully soft-analytic plaintext-checking oracle attacks against ML-KEM using probabilistic side-channel information and belief propagation.

Education

M.Sc. Cryptology and Cybersecurity, Specialisation in Algebra, coding theory and Cryptology Oct 2024 – Jul 2026

Centre for Cybersecurity and Mathematical Cryptology, University of Bamenda, Cameroon

Full academic scholarship

- **Supervised by:** Julius Hermelink, Peter Schwabe and Emmanuel Fouotsa
- **Grade:** First Class Honours
- **Relevant Coursework:** post-quantum cryptography, Coding Theory, unconditionally secure cryptography
- **Thesis:** Fully Soft-Analytic Plaintext-Checking Oracle Attack on ML-KEM

B.Sc. in Mathematics

Oct 2021 – Jun 2024

University of Douala – Douala, Cameroon

- **Grade:** First Class Honours(Très-bien)
- **Speciality:** Algebra, Analysis and Geometry

Experience

Research Intern in Applied Cryptography

Jun 2026 – Present

Max Planck Institute for Security and Privacy (MPI-SP), Bochum, Germany

- Design and implemented (Python and Rust) a unified factor graph to improve belief-propagation-based plaintext-checking oracle attacks against ML-KEM.
- Improve query complexity and noise tolerance
- Extended the factor-graph framework to investigate masking countermeasures

Private Tutor

2024–2025

- Machine Learning (Python, Scikit-learn, TensorFlow)
- General Topology
- Differential Equations

Key Research Interests

Post-quantum cryptography, side-channel analysis, belief propagation, lattice-based cryptography, Code-based cryptography, Zero-knowledge proofs

Academic Events

- **2026 School on Arithmetic, Geometry and Algorithms**, Addis Ababa, Ethiopia (August 2026). Selected participant; fully funded by the University of Groningen and the University of Amsterdam (ICTP-backed).

Grants and Awards

- **Fegù Academic Excellence Award (Highest Honours), 2024** – Union des Groupements des Elites Bandjoun (UGEB). Awarded outstanding national exam performance.
- **Undergraduate Class Top Student, 2024** – University of Douala.
- **JESAM Academic Excellence Award, 2021-** Mifi Department, Cameroon

Leadership and Service

Class Coordinator

Sept 2023 – Jun 2026

MSc Cohort 2024–2026 and BSc Cohort Sept 2023– Jun 2024

- Coordinated academic communication and class activities.

Key Skills

Programming languages: Python, C, Rust, Bash

Cryptography tools: SageMath, Macaulay2, OpenSSL

ML/Data: Pandas, Scikit-learn, Keras

Database technologies: MySQL, PostgreSQL

Typesetting: LaTeX

Languages

French: native; English: Fluent